Roberto André Mossi Milla

Computer Vision Engineer | Machine Learning | AI Systems

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Summary

Computer Vision and Machine Learning Engineer with experience building production-facing vision workflows, AI-enabled applications, and GPU-accelerated research simulations. Combines C++, Python, CUDA, PyTorch, YOLO, OCR, multimodal AI, and full-stack engineering to take vision systems from data ingestion and inference through human validation and operational use.

Experience

Gannon University — **Aug. 2024 – May 2025**
Computer Vision Engineer — Industry-Sponsored Project — *Erie, PA*

- Built components of a real-time industrial traceability system using YOLO-based object detection, event-driven services, TimescaleDB, Docker, and full-stack monitoring dashboards.
- Integrated vision-event data with persistent storage and operational interfaces, enabling downstream production workflows to review and act on detection results.
- Developed data and monitoring components that improved visibility into real-time computer-vision events across interconnected services.

Gannon University — **Aug. 2024 – May 2025**
Research Assistant — Evolutionary Artificial Intelligence — *Erie, PA*

- Conducted faculty-supervised research on autonomous multi-agent learning using NeuroEvolution of Augmenting Topologies (NEAT), investigating emergent behavior in dynamic simulated environments.
- Developed GPU-accelerated reinforcement-learning simulations in Unreal Engine using C++, CUDA, and PyTorch.
- Co-authored peer-reviewed research published through the American Society for Engineering Education (ASEE) and presented findings at a Sigma Xi conference hosted by Penn State University.

Ingeniería Digital — **May 2026 – Present**
Software Engineer (AI Systems) — *Honduras*

- Developed an AI vehicle-damage assessment workflow combining multimodal AI, computer vision, VIN-history retrieval, image analysis, and human review to support insurance valuation and claims decisions.
- Implemented Konva-based visual annotation tools that allow users to inspect, correct, and validate AI-assisted damage assessments.
- Built Playwright data-acquisition services to retrieve, normalize, and validate vehicle information from external sources for AI-assisted decision workflows.
- Developed an AI-assisted medical documentation platform integrating Amazon Textract OCR, OpenAI APIs, cloud storage, and structured document generation for healthcare professionals.
- Designed full-stack services using React, Next.js, TypeScript, PostgreSQL, Auth0, and AWS S3 to support secure document review, versioning, and workflow execution.

NOIR — **Jan. 2026 – Present**
Founder & Software Engineer (In Development) — *Remote*

- Designing a cross-platform movie discovery platform using SwiftUI and Rust services built around Axum, PostgreSQL, Redis, authentication, and cloud-based infrastructure.
- Architecting metadata aggregation and search workflows; planned AI assistance is limited to focused movie-discovery queries.

Dinant — **Jun. 2023 – Jul. 2023**
Full-Stack Developer — *Honduras*

- Developed database-backed business workflow features using Oracle SQL, JavaScript, HTML, and CSS, contributing to data modeling, reporting, filtering, and application delivery.

Technical Skills

Computer Vision & ML: YOLO, PyTorch, CUDA, computer vision, multimodal AI, OCR, image analysis, object detection, reinforcement learning, NEAT
Languages: Python, C++, Rust, TypeScript, JavaScript, SQL, Java, Bash
Systems & Data: Docker, Linux, TimescaleDB, PostgreSQL, Redis, REST APIs, event-driven services, AWS S3, Amazon Textract
Application Engineering: React, Next.js, Node.js, Playwright, Konva, Auth0, GitHub Actions, CI/CD

Education & Recognition

Gannon University — **2025**
B.S. Computer Systems Engineering — *Erie, PA*

- Peer-reviewed publication with ASEE; research presentation at Sigma Xi.